DSA CA1 Individual Project Report

Thomas Malinovsky - 24378831

Repo: https://github.com/ImDarkTom/DSACA1

Chosen Theme

The theme I have chosen from the Smart Docklands Pilot Call Themes is "Urban Greening/Biodiversity", specifically "Integrate data-driven tools to monitor and manage biodiversity, assessing how these insights can inform local and national climate action plans."

Description

The application will allow users to input and track a list of green areas and the biodiversity within those green areas. Each area will be part of a category (e.g. 'park', 'forest', 'marsh', etc.) and will track the specific flora and fauna within that area.

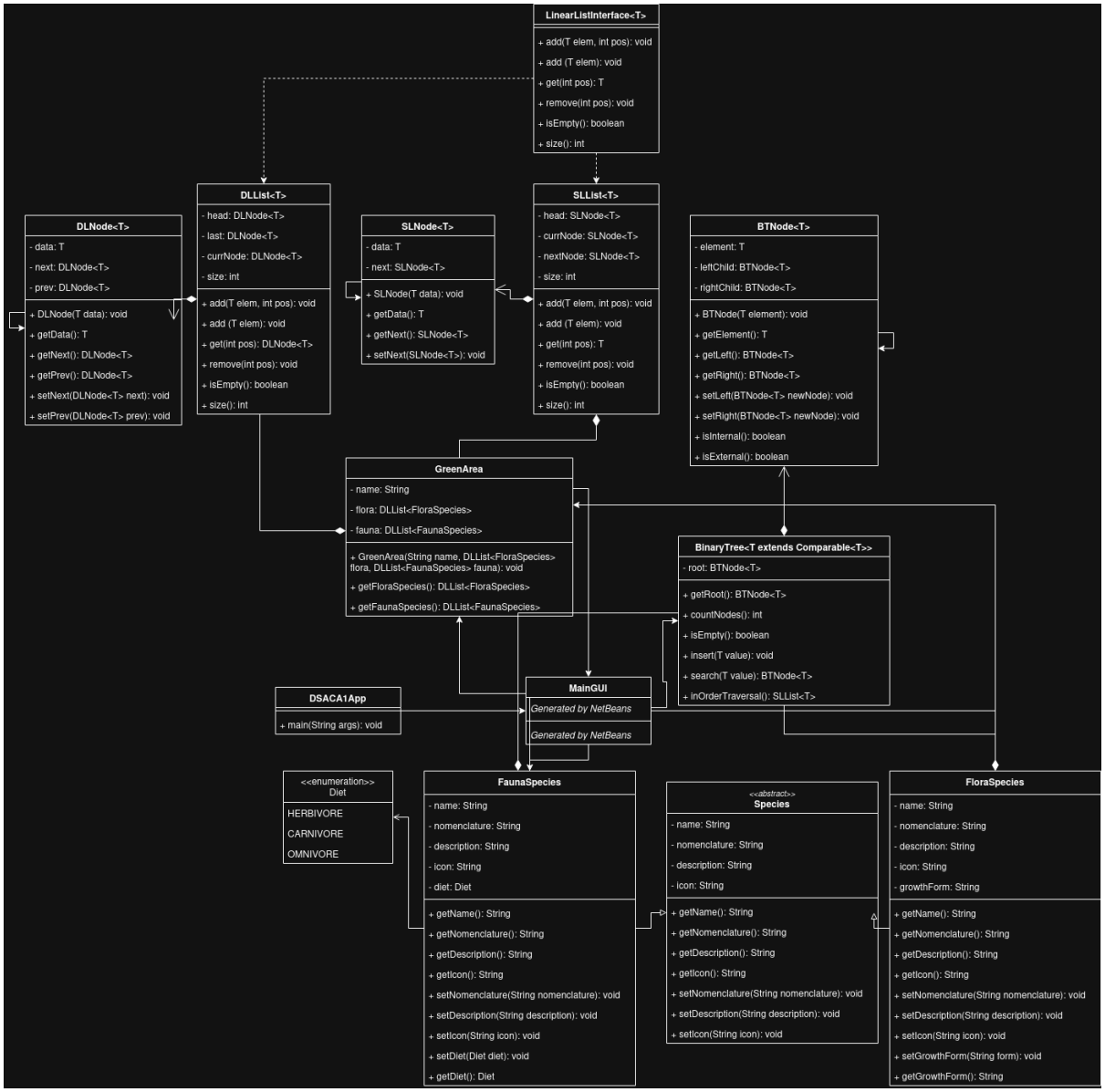
We will have a centralised global registry of flora and fauna species which will allow us to track and keep consistency for our flora/fauna. Each area will be able to add items from this registry to the area.

Problem being solved

The problem being solved is that is it difficult to plan and gain insights into national biodiversity, this app aims to address the issue of keeping track of biodiversity, as it will allow us to keep track of what flora and fauna is within each green area. Since we have a centralised list of all flora / fauna species, it will also mean that we will be able to see which areas have similar biodiversity, areas with dissimilar/low biodiversity, as well as areas that might be of concern.

How it fits the theme

It directly matches the theme of "Urban Greening/Biodiversity" as it provides a structured and data-driven way of managing biodiversity across different green areas around the country. Since we allow the user to record and categorise different flora and fauna found within these green areas in a centralised dataset, it will allow us to use that

dataset to track ecological patterns and use the insights to create action plans by highlighting potential issues in green areas.

Class Diagram